NODI / PIN-TO-PIN WIRING / REV B

25 October 2026 | Hardware untested | Cut lengths are allowances; trim after physical dry fit.

SOURCE PIN	NET / COLOR	DESTINATION / GAUGE / CUT LENGTH
W01 J1.6	+5V_RAW / red	F1.1 20 AWG 300 mm
W02 J1.5	GND / black	TP2.1 20 AWG 300 mm
W03 F1.2	+5V_FUSED / red	TP1.1 20 AWG 60 mm
W04 TP1.1	+5V_FUSED / red	J2.2 24 AWG 80 mm
W05 TP2.1	GND / black	J2.1 24 AWG 80 mm
W06 R1.2	YAW_SIG / purple	J2.3 26 AWG 80 mm
W07 TP1.1	+5V_FUSED / red	J4.2 24 AWG 90 mm
W08 TP2.1	GND / black	J4.1 24 AWG 90 mm
W09 R2.2	PITCH_SIG / green	J4.3 26 AWG 90 mm
W10 U1.36	+3V3 / orange	J4.5 28 AWG 90 mm
W11 TP2.1	GND / black	J4.4 28 AWG 90 mm
W12 U1.6	I2C_SDA / blue	J4.6 28 AWG 90 mm
W13 U1.7	I2C_SCL / yellow	J4.7 28 AWG 90 mm
W14 J4.1	GND / black	J3.1 24 AWG 180 mm FLEX
W15 J4.2	+5V_FUSED / red	J3.2 24 AWG 180 mm FLEX
W16 J4.3	PITCH_SIG / green	J3.3 26 AWG 180 mm FLEX
W17 J4.4	GND / black	U3.QA1 28 AWG 180 mm FLEX
W18 J4.5	+3V3 / orange	U3.QA2 28 AWG 180 mm FLEX

W14-W20: >=80mm free moving loop. Pin numbers are circuit/cavity IDs; see connector rules on page3.

Local passives and IC ties: NODI_System_RevB.pdf. Detailed wire notes: ../wiring_schedule.csv.

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SOURCE PIN	NET / COLOR	DESTINATION / GAUGE / CUT LENGTH
W19 J4.6	I2C_SDA / blue	U3.QA3 28 AWG 180 mm FLEX
W20 J4.7	I2C_SCL / yellow	U3.QA4 28 AWG 180 mm FLEX
W21 U1.36	+3V3 / orange	U4.1 28 AWG 70 mm
W22 TP2.1	GND / black	U4.2 28 AWG 70 mm
W23 U1.6	I2C_SDA / blue	U4.4 28 AWG 70 mm
W24 U1.7	I2C_SCL / yellow	U4.3 28 AWG 70 mm
W25 U1.20	BUTTON_N / white	SW1.A1 28 AWG 100 mm
W26 TP2.1	GND / black	SW1.B1 28 AWG 100 mm
W27 U1.4	YAW_PWM_3V3 / purple	U2.2 28 AWG 80 mm
W28 U1.5	PITCH_PWM_3V3 / green	U2.5 28 AWG 80 mm
W29 U1.21	AUDIO_PWM_3V3 / brown	R5.1 28 AWG 80 mm
W30 TP1.1	+5V_FUSED / red	BZ1.1 28 AWG 70 mm
W31 Q1.3	PIEZO_SW / brown	BZ1.2 28 AWG 70 mm
W32 TP1.1	+5V_FUSED / red	U2.14 24 AWG 40 mm
W33 TP2.1	GND / black	U2.7 24 AWG 40 mm
W34 TP1.1	+5V_FUSED / red	D1.2 24 AWG 40 mm
W35 D1.1	VSYS_EXT / red	U1.39 24 AWG 90 mm
W36 TP2.1	GND / black	U1.3 24 AWG 90 mm

W14-W20: >=80mm free moving loop. Pin numbers are circuit/cavity IDs; see connector rules on page3.

Local passives and IC ties: NODI_System_RevB.pdf. Detailed wire notes: ../wiring_schedule.csv.

NODI / CONNECTOR VIEWS AND FABRICATION NOTES

J4 / seven-way fixed-base disconnect

JST B7B-XH-A header + XHP-7 housing + SXH-001T-P0.6 contacts. Match molded cavity1.

Mating-face project cavity sequence: 1 servoGND | 2 +5V | 3 pitchSIG | 4 OLEDGND | 5 +3V3 | 6 SDA | 7 SCL.

J2 / J3 / servo plug mating faces

Project numbering: [1 GND] [2 +5V] [3 SIG]. Center contact is supply.

Verify actual servo colors and continuity; vendor JR plugs have no assumed molded integer numbering.

U3 / STEMMA QT

Physical connector contacts: [1 GND] [2 VIN=3.3V] [3 SDA] [4 SCL].

QT wire colors: black GND; red VIN (3.3V in NODI); blue SDA; yellow SCL. Other module header pins NC.

J1 / USB-C input board component-side view

USB socket LEFT; six-pad header TOP to BOTTOM: 6 VBUS, 5 GND, 4 CC1, 3 D-, 2 D+, 1 CC2.

Current BOB-15100 already contains two5.1k Rd. Do not fit parallel external CC resistors.

SW1 / all four physical legs

Project labels A1/A2 identify one internally common pair; B1/B2 the other. Press joins A to B.

Wire one A leg and one B leg. Do not use two legs from the same pair; verify pairs using continuity mode.

Construction scope

TP1/TP2 are covered solder-star joints. Use separate motor returns; do not route servo power through Pico.

J4 is an insulated tethered disconnect: no fit/retention claim in current CAD. Input pod is outside the base.

Before powering

Continuity-check every CSV wire; inspect shorts, polarity, fuse, cathode band, transistor pinout and module revision.

Power current-limited without servos, then center with horns removed. Sweep all combined poses before live motion.

Connector views are functional project diagrams, not scaled machining drawings. Wire-entry views mirror mating faces.

No hardware build, electrical measurement or harness fatigue test has been performed.

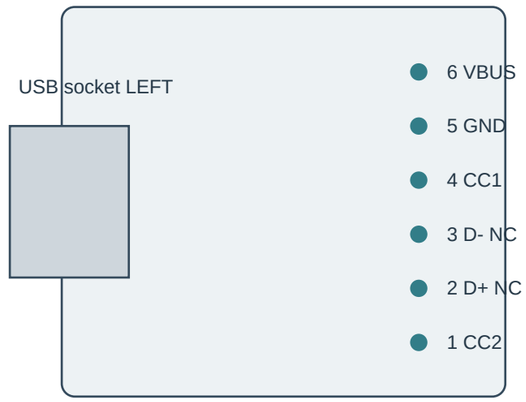
NODI / PHYSICAL CONNECTOR PIN MAPS

Functional views: not scaled drawings. Match molded pad1 or project marks; wire-entry views mirror mating faces.

U1 / PICO / USB UP

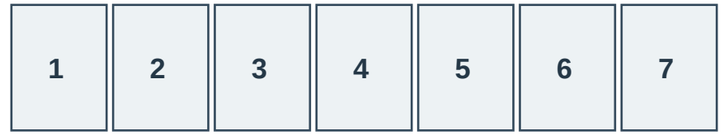
Other GND pads are internally common; pin3 is external return.

J1 / COMPONENT SIDE



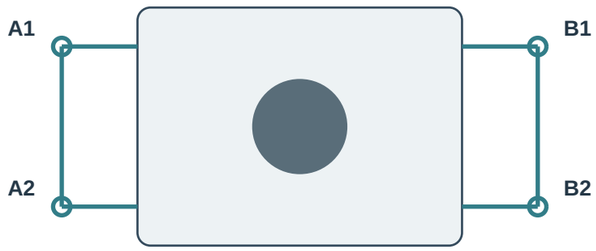
J4 / MATING OPENINGS TOWARD YOU

Orient molded cavity1 at LEFT. Back/wire-entry view is mirrored.



GND | +5V | SIG | GND | +3V3 | SDA | SCL

SW1 / TOP VIEW / VERIFY PAIRS BY CONTINUITY



W25 to A1; W26 to B1. A2/B2 require no additional wires.

J2 / J3 / SERVO MATING FACE

Mark GND on housing; confirm the actual servo lead colors.

The pin maps supplement the circuit: supply voltage and cavity continuity must be checked before energizing.

J4 is an unmodeled, separately insulated/tethered disconnect; XH2.50mm pitch is not standard2.54mm perfboard pitch.